2.7 Derivatives of $\sin x$, $\cos x$, e^x and $\ln x$
2.8 The Product Rule

Ex 1:

Let f be the function given by $f(x) = x^3 + 5x$. For what value of x in the closed interval $[1,3]$ does the instantaneous rate of change of f equal the average rate of change of f on that interval?

(A) $\sqrt{\frac{7}{3}}$ (B) $\sqrt{\frac{13}{3}}$ (C) $\sqrt{5}$ (D) $\sqrt{6}$ (E) $\sqrt{\frac{19}{3}}$

$$f'(x) = \frac{f(3)-f(1)}{3-1}$$
$$3x^2+5 = \frac{42-6}{3-1}$$
$$x = \pm\sqrt{\frac{13}{2}}$$

$-\sqrt{\frac{13}{2}}$ *is extraneous since the interval is [1,3]*

Ex 2: Fill in the blank.

a) A function, $f(x)$, is continuous at $x = a$ if $f(a) = \lim_{x\to a} f(x)$.

b) A function is differentiable at $x = a$ if $f(x)$ is continuous at $x = a$ and $f'(a) = \lim_{x\to a} f'(x)$.

Ex 3:

At $x = 3$, the function given by $f(x) = \begin{cases} x^2, & x < 3 \\ 6x - 9, & x \geq 3 \end{cases}$ is

(A) undefined.
(B) continuous but not differentiable.
(C) differentiable but not continuous.
(D) neither continuous nor differentiable.
(E) both continuous and differentiable.

$f(3) = \lim_{x\to 3} f(x) = 9$, $\therefore f$ is continuous at $x = 3$.

$$f'(x) = \begin{cases} 2x, & x < 3 \\ 6, & x \geq 3 \end{cases}$$

Since $f'(3) = \lim_{x\to 3} f'(x) = 6$ and f is continuous at $x = 3$, f is also differentiable at $x = 3$.

Ex 4: Differentiate.

a) $f(x) = \begin{cases} x^2, & x \leq 1 \\ 1 + 2x, & x > 1 \end{cases}$

$\lim_{x\to 1} f(x)$ DNE, f is NOT continuous at $x = 1$ and $\therefore$ not differentiable at $x = 1$.

$$f'(x) = \begin{cases} 2x, & x < 1 \\ 2, & x > 1 \end{cases}$$

b) $f(x) = |x + 4|$

$$f(x) = \begin{cases} x + 4, & x \geq -4 \\ -(x+4), & x < -4 \end{cases}$$

$f(-4) = \lim_{x\to -4} f(x) = 0$, $\therefore f$ is continuous at $x = -4$.

$$f'(x) = \begin{cases} 1, & x \geq -4 \\ -1, & x < 4 \end{cases}$$

Since $\lim_{x\to -4} f'(x)$ DNE, f is not differentiable at $x = -4$.

Ex 5: Find the slope of $f(x) = |x + 4|$ at the indicated x –value, or explain why it does not exist.

a) $x = 0$

$f'(0) = 1$

b) $x = -5$

$f'(0) = -1$

c) $x = -4$

Since $\lim_{x\to -4} f'(x)$ DNE, f is not differentiable at $x = -4$ and $f'(0)$ DNE.

Ex 6: Let f be the piecewise-linear function defined. Which of the following statements are true?

$$f(x) = \begin{cases} 2x - 2 & for\ x < 3 \\ 2x - 4 & for\ x \geq 3 \end{cases}$$

I. $\lim_{h \to 0^-} \frac{f(3+h) - f(3)}{h} = 2$

II. $\lim_{h \to 0^+} \frac{f(3+h) - f(3)}{h} = 2$

III. $f'(3) = 2$

(A) None

(B) II only

(C) I and II only

(D) I, II and III

$\lim_{x \to 3} f(x)$ DNE, f is NOT continuous at $x = 3$ and $\therefore$ not differentiable at $x = 3$.

Ex 7: Find a and b so that $f(x) = \begin{cases} 2x - x^2, & x \leq 1 \\ x^2 + ax + b, & x > 1 \end{cases}$ is differentiable.

$f(1) = \lim_{x \to 1} f(x)$

$1 = 1 + a + b$

$a + b = 0$

$f'(x) = \begin{cases} 2 - 2x, & x \leq 1 \\ 2x + a, & x > 1 \end{cases}$

$f'(1) = \lim_{x \to 1} f'(x)$

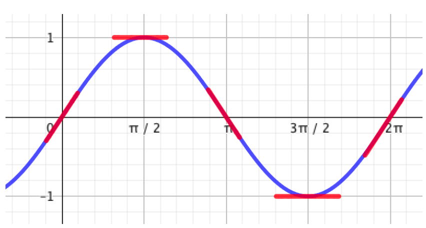
$2 - 2 = 2 + a$

$a = -2$

$b = 2$

Ex 8: Below are the graph of $f(x) = \sin x$.

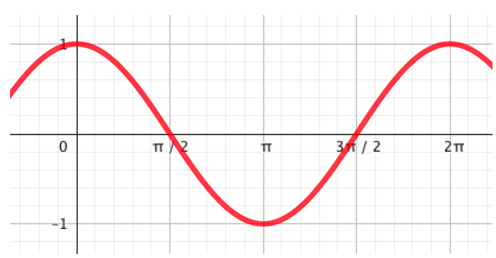
a) Use the graph to fill in the table.



x	$f'(x)$
0	1
$\frac{\pi}{2}$	0
π	-1
$\frac{3\pi}{2}$	0
2π	1

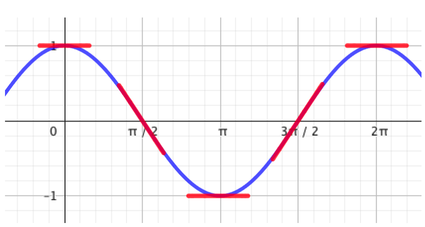
b) Plot the points $(x, f'(x))$ and connect the dots. This is the graph of the derivative of $f(x) = \sin x$. Look familiar?

$f'(x) = \cos x$



Ex 9: Below are the graph of $g(x) = \cos x$.

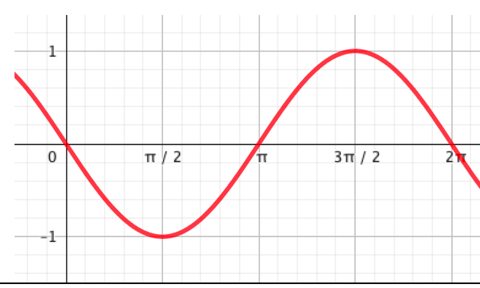
a) Use the graph to fill in the table.



x	$g'(x)$
0	0
$\frac{\pi}{2}$	-1
π	0
$\frac{3\pi}{2}$	1
2π	0

b) Plot the points $(x, g'(x))$ and connect the dots. This is the graph of the derivative of $g(x) = \cos x$. Look familiar?

$g'(x) = -\sin x$



Differentiation Rules – Sine and Cosine

$$\frac{d}{dx}[\sin x] = \cos x \qquad \frac{d}{dx}[\cos x] = -\sin x$$

Ex 10: Differentiate: $f(x) = 3\cos x + 2$

$$f'(x) = -3\sin x$$

Ex 11: What is the slope of the tangent line to the graph of $f(x) = 3\cos x + 2$ at the point $x = \frac{3\pi}{2}$?

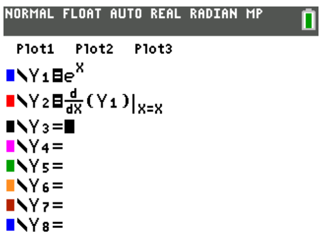
$$f'\left(\frac{3\pi}{2}\right) = -3\sin\left(\frac{3\pi}{2}\right) = 3$$

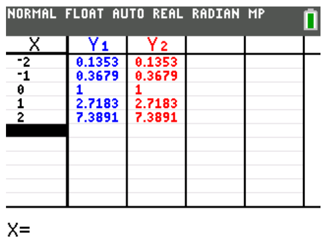
$$f\left(\frac{3\pi}{2}\right) = 2$$

$$y - 2 = 3\left(x - \frac{3\pi}{2}\right)$$

Ex 12: Consider the function $h(x) = e^x$.

a) Use a graphing utility to fill in the table.





NORMAL FLOAT AUTO REAL RADIAN MP

X	Y1	Y2
-2	0.1353	0.1353
-1	0.3679	0.3679
0	1	1
1	2.7183	2.7183
2	7.3891	7.3891

X=

b) Compare the $h(x)$ and $h'(x)$ values in the table. What do you notice?

The $h(x)$ and $h'(x)$ values are the same!

c) $h'(x) = e^x$

Differentiation Rules – Exponential Functions

$$\frac{d}{dx}[e^x] = e^x \qquad \frac{d}{dx}[a^x] = a^x \ln a$$

Ex 13: Differentiate: $f(x) = 5e^x + \sqrt{x}$

$$f'(x) = 5e^x + \frac{1}{2\sqrt{x}}$$

Differentiation Rules – Logarithmic Functions

$$\frac{d}{dx}[\ln x] = \frac{1}{x},\ x > 0 \qquad \frac{d}{dx}[\log_a x] = \frac{1}{x\ln a},\ x > 0$$

Ex 14: Differentiate: $g(x) = 7\sin x + \ln x - 2 \cdot 5^x + \frac{8}{x^3}$

$$g'(x) = 7\cos x + \frac{1}{x} - 2 \cdot 5^x \ln 5 - \frac{24}{x^4}$$

Product Rule

$$\frac{d}{dx}(f(x) \cdot g(x)) = f(x) \cdot g'(x) + f'(x) \cdot g(x)$$

An easy way to remember the product rule: derivative sandwich!

Ex 15: Find the derivative.

a) $y = (5x+1)\ln x$

$$y' = (5x+1)\cdot\frac{1}{x} + 5\cdot\ln x$$

or

$$y' = 5 + \frac{1}{x} + 5\ln x$$

b) $f(x) = x^5\cos x$

$$f'(x) = -x^5\sin x + 5x^4\cos x$$

Not all products must be differentiated with the product rule. Sometimes it is easier or quicker to REWRITE first!

Ex 16: Which equations **must** be differentiated with the product rule?

a) $f(x) = 4\sin x$

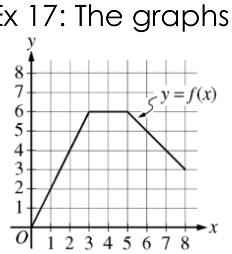
b) $y = xe^x$

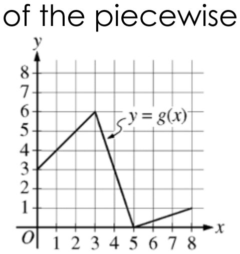
c) $y = (x+1)(x^2-3)$

d) $g(x) = 2x^3$

e) $y = 2\sin x\cos x$

Ex 17: The graphs of the piecewise linear functions f and g are shown.





If $h(x) = f(x)\cdot g(x)$, find $h'(7)$ or state that it does not exist.

$$h'(x) = f(x)\cdot g'(x) + f'(x)\cdot g(x)$$
$$h'(7) = f(7)\cdot g'(7) + f'(7)\cdot g(7)$$
$$h'(7) = (4)\left(\frac{1}{3}\right) + (-1)\left(\frac{2}{3}\right) = \frac{2}{3}$$

Ex 18: The functions $f(x)$ and $g(x)$ are differentiable. The table gives values of the functions and their first derivatives at selected values of x.

x	0	2	4	7
$f(x)$	10	7	4	5
$f'(x)$	$\frac{3}{2}$	-8	3	6
$g(x)$	1	2	-3	0
$g'(x)$	5	4	2	8

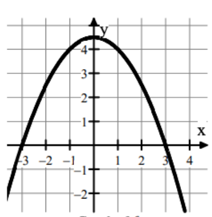
a) If $h(x) = f(x)\cdot g(x)$, find $h'(0)$ or state that it does not exist.

$$h'(x) = f(x)\cdot g'(x) + f'(x)\cdot g(x)$$
$$h'(0) = f(0)\cdot g'(0) + f'(0)\cdot g(0)$$
$$h'(0) = (10)(5) + \left(\frac{3}{2}\right)(1) = 51.5$$

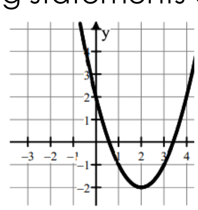
b) Write a tangent line to $h(x)$ at $x = 0$.

$$h(0) = (10)(1) = 10$$
$$y - 10 = 51.5(x-0)$$

Ex 19: The graphs of two differentiable functions f and g are shown below. If $h(x) = f(x)\cdot g(x)$, which of the following statements about $h'(3)$ are true?



Graph of f



Graph of g

a) $h'(3) < 0$

b) $h'(3) = 0$

c) $h'(3) > 0$

d) $h'(3)$ is undefined

e) There is not enough information to conclude anything about $h'(3)$.

$$h'(x) = f(x)\cdot g'(x) + f'(x)\cdot g(x)$$
$$h'(3) = f(3)\cdot g'(3) + f'(3)\cdot g(3)$$
$$h'(3) = (0)(+) + (-)(-) > 0$$